

11

Exploratory Factor Analysis

CHAPTER OBJECTIVES

- Learn the nature of exploratory factor analysis (EFA) and why, as a scientist, you may wish to perform one.
- Distinguish between EFA and PCA, and understand why they should not be likened to one another and are quite distinct statistical methods.
- Appreciate the fact that EFA has several philosophical pitfalls that can make interpreting findings from the approach relatively subjective.
- Understand that solutions in factor analysis are not invariant to the number of factors selected for extraction.
- Understand the nature of factor rotation and why rotating factors is not “cheating” or “fudging data” so to speak.
- How to perform a basic EFA in Python and interpret results.

As was the case for principal components analysis (PCA) of the previous chapter, **exploratory factor analysis** (EFA) can be seen as a **dimension reduction** technique in which a researcher begins with several observed variables and wishes to “uncover,” in a manner of speaking, dimensions that underlie the observed data. We place “uncover” in quotes here for the good reason that we must be cautious in how we interpret this word in relation to factor analysis, as we also had to be cautious in our discussions of PCA of the previous chapter. As we will see throughout this chapter, factor analysis does not “find” factors in one’s data in an objective concrete sense. In fact, we will see, whether or not factor analysis results in a viable solution often depends on very subjective decisions made by the researcher. Hence, “finding factors” in factor analysis is quite unlike that of the archaeological dig we referred to in which an explorer discovers remnants of past civilizations. In factor analysis, **what you “find” often depends on assumptions and inputs you a priori decide for the analysis.** This point cannot be overstated.

By the end of the chapter, it is hoped that you will appreciate both the benefits but also the drawbacks to using and interpreting an EFA. Without understanding factor analysis, just as with any statistical method, you are more likely to misuse and

misinterpret its findings. Though this is also true for methods such as ANOVA and regression, the situation is even more dire with factor analysis, as we will discuss. It is exceptionally difficult for a statistical consultant to recommend that you perform factor analysis unless they are intimately familiar with your research area. That is, when it comes to factor analysis, **substantive knowledge of the domain of interest is paramount**. Do you believe **latent variables** give rise to the observed variables you have measured? If so, then EFA may be an option. Be forewarned that factor analysis is an immense and very technical subject and hence our treatment here is very cursory. For a detailed account of all of its complexities, consult Mulaik (2009).

Before surveying EFA in some detail, it is key at the outset that you do not equate factor analysis with the method of the previous chapter, that of PCA. **EFA, while sharing some similarities with PCA, is not an equivalent method**, and there are very important differences between the two procedures. A failure to understand the differences between these two procedures can lead to a misuse and misinterpretation of either or both.

11.1 The Common Factor Analysis Model

As mentioned, factor analysis is a huge topic spanning many technical articles, chapters, and books. Of all the statistical methods featured in this book, factor analysis, for good or bad, is probably the one that has the most **complexity** associated with it. Why is this the case? It is so because there are so many options and inputs to factor analysis that to truly become an efficient expert at it, you should have at least some understanding of these many issues before using the technique. In this book, we survey only one approach to factor analysis, which is undoubtedly the most common. This is referred to as the **common factor analysis model**, and is given by the following:

$$\mathbf{x} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{f} + \boldsymbol{\epsilon}$$

where $\mathbf{x}$ is a vector of **observed random variables**. These are variables the researcher has collected and would like to subject to the factor analysis (you might call them the “candidate” variables). The number of variables in this vector can range from very few to a very large number, depending on the complexity of the research problem. To use a previous example, suppose a researcher’s observed variables include quantitative, verbal, and analytical abilities as measured on some psychometric test. These are regarded as “observed” variables because we are able to obtain numerical measurements on them, and hence they are considered readily available to the researcher. In the language of factor analysis, they can be “seen” by the score on the psychometric test. In factor analysis language, this is different from an **unobserved** or “**latent**” variable, which is a variable that is regarded as not readily observable and hence must be “inferred” in a manner of speaking from other more observable variables. In the example we have used throughout the book, perhaps quantitative, verbal, and analytical, taken as a set, are indicators of an underlying construct or more complete variable, perhaps that of IQ. The primary rationale for performing a factor analysis is because the analyst hypothesizes or believes that underlying such observed variables is a **latent structure** that cannot be seen but rather is “responsible” for generating the correlations among the observed variables. Hence, if a researcher does not have this hypothesis in mind, then performing factor

analysis may be ill-advised. Factor analysis focuses on the covariances and correlations among variables. If the researcher simply wants to extract orthogonal sources of variance, and hence simply reduce dimensionality without the assumption of underlying latency giving rise to observed variables, then the principal components of the previous chapter would usually be the preferred analytical strategy.

Returning to the factor analysis model, μ is a vector of means for the random variables contained in $\mathbf{x}$, and Λ is a matrix of factor loadings applied to the factors in $\mathbf{f}$. Hence, $\mathbf{f}$ in this case is a vector of **unobservable factors** that are weighted by the loadings in Λ . This idea should sound very familiar, because it is somewhat analogous to regression analysis. But how is factor analysis different from regression? In regression analysis, recall that instead of $\mathbf{f}$, an unobservable vector, the regression model instead contained $\mathbf{x}$, that is, **observable variables**. Each of these variables were, analogous to Λ , weighted by β , the vector of regression weights. Hence, we see the primary distinction between factor analysis and regression analysis. In regression analysis, nowhere in the model was there hypothesized to be a latent term, other than the error term ϵ , which can always be considered latent (i.e. errors are usually considered “unobservable”). Otherwise, all predictor variables are considered known and measurable without error in the classic regression model. In the factor analysis model, however, we are hypothesizing observed variables in $\mathbf{x}$ as a function in part of unobservable variables called “factors” in $\mathbf{f}$.

As mentioned, the factor analysis model contains a vector of errors, ϵ . These are generally known as **specific factors** in factor analysis and account for the variation unaccounted for by the common factors in the model. That is, anything that cannot be explained by $\mathbf{f}$ winds up in the error term. This is somewhat analogous to regression analysis where what cannot be accounted for by predictors ends up in the error term. In a more advanced study of factor analysis, some theorists will further differentiate specific factors into further components of variance, but doing so is beyond the scope of this book. For details, see Mulaik (2009).

Hence, we can see that the factor analysis model is, in a sense, closely analogous to the regression model, but with the caveat that other than the error term, no other unobservable variables appear in the regression model. In both cases, however, both models attempt to account for a systematic component of the data vs. an unsystematic one. But factor analysis does so by theorizing **unobservable latent dimensions**, and hence right from the start is more controversial than the traditional regression model.

11.2 Factor Analysis as a Reproduction of the Covariance Matrix

Though we have seen that the factor analysis model can be correctly interpreted as observed variables being a function of unobserved variables, mathematically it can be shown that when the factor analysis model fits the data well, the model itself implies a **reconstruction of the covariance (or correlation) matrix**. Though it is well beyond the scope of this book to justify why this is the case (see Denis, 2021, for details), it turns out that the factor model implies that the covariance matrix of observed variables can be reconstructed by the following:

$$\Sigma = \Lambda\Lambda' + \Psi$$

where Σ is the covariance of observed variables, and Λ are the factor loadings we just discussed and surveyed. The matrix Ψ is one of specific variances. Hence, what the equation says is that the covariance matrix can be written as a function of “squared loadings,” where $\Lambda\Lambda'$ is the “squared” (i.e. we are multiplying by the transpose and not technically squaring, but it is helpful to think of it as squaring in this case) component represented by the loading matrix multiplied by its transpose, in addition to the matrix of specific variances. If the factor model is a good fit to the data, then it follows that the covariance matrix should likewise be reproducible by $\Lambda\Lambda' + \Psi$. The sum of such “squared” factor loadings for a variable across the factor solution is what is known as a **communality**, and, as the name suggests, tells us the degree to which the variance in the variable is “explained” by the factor solution. The balance of the communality, that is, the variance **unexplained** by the factor solution for each variable, is the **specific variance** associated with that given variable. These quantities will make more sense later when we see them estimated in our software examples. Notice that it is the variance in the observed variable we are seeking to explain by the factor solution, and not the other way around. In the factor model, the observed variable is theorized to be a function of latent variables. Given all this, we can also interpret the goal of factor analysis in this slightly more technical yet simplified way; **factor analysis attempts to estimate loadings in such a way that they maximize the reproduction of the covariance matrix**. In the case of a correlation matrix, which recall is simply a standardized covariance matrix, the goal is the same. Regardless of the matrix, factor analysis is, “behind the scenes,” trying to “reproduce” it.

This idea of “reproducing” a function of the variables may sound familiar. We can again relate it to regression. Recall that the goal of least-squares regression is to minimize the sum of squared errors in prediction. What this translates into as well is maximizing the correlation between observed and predicted values on the response variable. Hence, when pondering the question of how well a regression model fits, one asks the question, “**How well does it reproduce a function of the observed variables?**” The extent that it does is the extent to which the model is well-fitting. This is a similar idea as in factor analysis, where we are hoping the model regenerates the observed covariance matrix of variables. Recall that models, by their nature, are in the business of hypothesizing a structure to observed data. More generally, this is precisely what mathematical modeling (of which statistical and “decision sciences” modeling is a special case) is all about, that of describing empirical data by an abstract mathematical equation or algorithm of sorts.

11.3 Observed vs. Latent Variables: Philosophical Considerations

Having introduced this idea of “latent” or “unobservable” variables, the insightful reader at this point should not be anywhere close to at ease or satisfied with the definitions given earlier. In fact, he or she should be quite troubled by them! Being critical and unsatisfied with trivial definitions is a first sign of intellectual depth and curiosity, which is a great thing! The question of whether an observed vs. latent distinction is “real” is definitely a good one, and we discuss the matter now in an attempt to resolve it, even if imperfectly.

At first glance, the idea of latent variables even “existing” may seem silly. That is, what does it mean to have a variable that “gives rise” in this sense to other more observable variables? Indeed, the idea of hidden variables or variables that lie below the surface (maybe “limen”?) is philosophically problematic, but then again, so is the idea of “observed” variables just as problematic. Yes, even so-called “observed” variables are problematic too. How so? Allow me to explain using a simple example of a variable none of us would, at first, deem philosophically debatable or troublesome, that of **weight**. Is weight observed or latent? Most people would say it is observable, since one can step onto a bathroom scale, receive a measurement, and learn of one’s current weight. However, even in this simple procedure, there are numerous psychometric issues that arise that at first thought you may not have considered.

First, weight is a **construct**, no matter how much we would like to think it is not (just follow me on this for a second). Weight is an “idea” that we believe underlies numerous other more observable events such as the amount of pressure being applied to the scale, the instrumentation involved in constructing the physical weight scale, and all of the other components to the measurement of weight. Hence, the variable “weight” does not exist separately from our inference of it based on all of these other components that contribute to the construct. That is, nowhere in nature is there “natural” weight. Like any variable, it is a **human construction**, and we choose to designate it as a sum component of numerous more observable events or procedures. Hence, the measurement of weight involves a lot more than just “weight.” To be blunt, the thing you step on in the morning (you know, to ruin your day) is also involved in this thing called “weight.” **The way we measure weight has a lot to do with what weight is.**

Second, when you step on the bathroom scale, what you are receiving as a measurement of your weight is not your actual weight. Rather, it is an **estimate** of your actual **true weight**. Your true weight will forever be unknown. Why? Because there is understood to be **measurement error** in the estimation of your true weight. Stepping on the scale does not reveal the true score for weight, it simply serves as an estimate of what the true score may be. Psychometrics 101! This is also true for all variables you may consider, not only weight, but virtually anything. Hence, “true weight” is virtually **unknowable**. However, if your measurement instrument is a good one, then you will get a good approximation to the true weight. Though measurement error may be reduced, it is never, at least on a theoretical level, eliminated. Hence, in this regard, “true weight” is a “hidden” variable. Now, surely launching **SpaceX** into space does not involve such measurement error, does it? The answer is yes it does! Good scientists and engineers use instruments that minimize measurement error, but it can never be avoided, at least not on a theoretical level, which means that even if you can avoid it in most practical situations, there will always be the possibility that a new measurement will contain such an error. In events like space shuttle launches, measurement error is extremely small, but it exists nonetheless. Hence, we must concur that **measurement error exists in all variables, from physical to social ones. In some variables it exists more than others.**

Thirdly, and related to the point just made above, it needs to be recognized that between an object and our seeking to know it is a **measurement procedure**. The measurement procedure is usually made up of some instrument or other process by which we seek to assign a value to the object, or at least our estimate of it. Again, this

is true for whatever object we consider, even ones where you would never think such issues exist! For instance, the measurement of heart rate is even an example of this. The medical doctor wishes to know something called heart rate and uses an instrument of measurement to learn of this information. What is the patient's true heart rate? That value is subject to a **measurement process**, and the measurement process is subject to generating measurement error in the thing that is being measured. Hence, the stethoscope may provide one reading of heart rate, while another measurement process may provide another (at least in theory). In theory, we hope they agree, but they may not. The key point is between anything we seek to "know," there is a need to **measure** it. By way of analogy, you are using a measurement process to read the pages of this book. Your eyes, combined with the rest of your sensory organization are capturing a representation of what is on the page. How accurately is it measuring? You may think even if your eyes are "20/20" you are capturing the true value, but you would be wrong! You are capturing the value afforded by your current instrument of measurement. We must allow for the possibility that, in the future, what is 20/20 measurement today becomes 18/20 measurement tomorrow. What you are seeing on the page is your current "best shot" at capturing what is there. Hence, in line with our discussion, what is on the page is a **latent** object you are trying to detect.

The analogy need not stop here. We can be even more general. The world is filled with other constructs or **processes** we wish to capture. Consider the tennis great Rafael Nadal, who is well known as the best clay court tennis player of all time. Regardless, every time he steps out onto the court, he is **learning how to play clay court tennis**. In our language, he is attempting to accurately measure a process, an ideal that is currently unknown to him, similar in some ways to your bathroom scale attempting to measure something unknowable. The key idea is to realize and understand that virtually all processes, variables, and otherwise "things" are unknowable in this regard, and in this sense, are latent, and not "observable" as they may at first glance appear.

We can safely conclude then that **all variables are latent, but some are more easily and precisely measured than others**. That is, in any measurement or process, we are seeking to know the true variable or "thing" under inspection. In some cases, such as weight, that approximation one may argue is quite close and requires a relatively simple measurement instrument. In other cases, such as IQ, the measurement is likely to be more complex and may require several other variables in order to "**triangulate**" on the variable that is being sought. At least, that is the argument. Of course, what defines correct measurement is rather arbitrary and not necessarily related to measurement error. That is, there is nothing preventing us from designating IQ as simply the number of finger taps on a desk one can accomplish in a one-minute time interval. Foolish and simple as such a method would be to capture or assess such a complex construct, it remains that how we choose to measure something is not only a function of the degree of measurement error that may exist, but is first and foremost an issue of **validity**. Hence, **validity in measurement** is even more important. Is measuring beats-per-minute of a heart a valid measurement of heart-rate? As a society, we generally agree that it is. Of course, this example is extreme (we are at the point of forcing these philosophical notions to their limits), but in essence is no different in nature than our agreement (or disagreement) as to what intelligence is (or is not). Ideally, we first agree that we are measuring the correct thing and then we attempt to

measure that thing with as little measurement error as possible using the **very best instruments currently available in line with how we operationalize the construct**. But we must also allow for the possibility that future instruments will reveal something different! Hence, in all of these cases, spanning the full spectrum, one may consider at an operational level for all variables to be **latent**, not observed. When we designate a variable as “observed” in factor analysis, we are simply considering it more readily available and as an indicator of a more latent or hidden process. Conclusion? Latent variables are those less readily available or observable than so-called “observed” variables. However, **all variables, every one, are in essence latent variables when subjected to proper scrutiny and philosophical (i.e. “intelligent” and “critical”) unwrapping.**

11.4 So, Why is Factor Analysis Controversial? The Philosophical Pitfalls of Factor Analysis

Having briefly surveyed the factor analysis model, it may seem as though our initial statement of it being potentially controversial is unfounded. After all, it looks like a pretty innocent model. So, what could be the problem? Why the controversy? The problem is that regardless of the solution obtained in factor analysis, **factor loadings are not uniquely defined**. That is, the loadings obtained in Λ when the factor model is estimated are not unique. The idea of “unique” in this capacity has different meanings, but for our purposes, it implies that the loadings obtained are not the only ones that could have been obtained for the factor analysis model to successfully regenerate the covariance or correlation matrix. For applied purposes, this has two major consequences for the user of factor analysis:

- **Estimated factor loadings for a particular factor are typically contingent upon how many other factors are estimated along with it.** That is, a given factor loading in an EFA solution is not a unique solution. The solution will depend on how many other factors are extracted along with that factor. This does not occur in PCA. In PCA, whether you extract one component or five, for example, the loading for the given component is **stable** and does not change.
- Given the above, the determination of the number of factors in data is a complex decision that many argue is a function just as much of “art” as it is of science. For example, if a factor analyst extracts a five-factor solution and finds evidence for two factors, that solution might change if the researcher re-analyzes the data extracting only two factors. Hence, the determination of how many factors “exist” in the data can be quite arbitrary in EFA. Again, this is not the case in PCA. In PCA, weights and loadings are stable across the number of components extracted or interpreted.
- A third, but not unrelated issue to the one mentioned just above, is that the loadings in EFA can be subjected to an **orthogonal rotation**, but thankfully still reproduce the covariance matrix. Hence, in this sense, we can say the loadings are unique, but only up to an orthogonal rotation. That is, a researcher can rotate the solution in EFA and still regenerate the same covariance or correlation matrix. Again, this would appear to make the **determination of the weights seem quite arbitrary.**

These issues should not be easily dismissed by the researcher as he or she continues on with performing a factor analysis on his or her variables. Neither do they suggest that factor analysis should not be used as a tool to help make sense of one's data. What these caveats do suggest, however, is that researchers should interpret factor-analytic solutions with caution and realize that the solution they obtained is not the same as the rigorous “**discovery**” as some would make it seem. That is, the researcher did not “find” a given factor solution in the sense that there is a concrete **existence** of the given number of factors. Rather, using the factor-analytic model, and all the assumptions along with it, the researcher obtained what he or she believes is a feasible and useful solution to help make sense of the data and make progress in the scientific field. That is all that has been found, nothing more. Again, to use the analogy of an archaeological dig, “finding” factors is not the same as digging into the ground and discovering an ancient skeleton relic. It is not even close to that! Researchers who perform factor analysis and those who consume its results need to be aware of this. Factors are not “discovered” in the sense that a vaccine for COVID-19 is invented, for instance. What is “found” in factor analysis is much more transient and subject to several inputs and assumptions. Truth be told, the identification of the COVID-19 virus under a microscope also depends on instrumentation, inputs, and assumptions, but that becomes a bit more of an academic discussion and distinction than a real one. At some point, we have to work with the instrumentation we have and assume what we are seeing “exists.” However, on a philosophical level (i.e. a “thinking about what we are doing” level, which is all “philosophical” means here), the “reality” of the virus under the microscope is subject to those same influences. Biologists usually do not have to confront such issues, but psychologists, if they are to do their work correctly and with care, regularly do (or at least, should). **Psychometrics** is a very exciting and challenging field. **The instrument of measurement is just as important as what is being measured.**

11.5 Exploratory Factor Analysis in Python

We now demonstrate an EFA on the same data on which we performed a PCA in the previous chapter. Recall the dataframe we built for the PCA. We rebuild it here:

```
import pandas as pd
import numpy as np

pca_data = np.array([[0, 5.90], [0.90, 5.40], [1.80, 4.40], [2.60,
4.60], [3.30, 3.50], [4.40, 3.70], [5.20, 2.80], [6.10, 2.80],
[6.50, 2.40], [7.40, 1.50]])

pca_data
Out[57]:
array([[0. , 5.9],
       [0.9, 5.4],
       [1.8, 4.4],
       [2.6, 4.6],
       [3.3, 3.5],
```



```
[4.4, 3.7],
[5.2, 2.8],
[6.1, 2.8],
[6.5, 2.4],
[7.4, 1.5]])
```

Recall that in PCA of the previous chapter, we sought a transformation such that the total variance in the variables is preserved. In EFA, we do not seek such a transformation, but rather wish to **fit the factor model** discussed earlier to the data, and in the process **estimate loadings** and **specific variances**.

Before we proceed with the factor analysis, we can conduct **Bartlett’s test of sphericity** to evaluate the null hypothesis that the correlation matrix is an **identity matrix**. If it is an identity matrix, then it makes little sense in performing the factor analysis, since it would indicate that we have zero pairwise correlations between variables (i.e. values of “0” on the off-diagonal positions of the matrix):

```
pip install factor_analyzer
import factor_analyzer
from factor_analyzer.factor_analyzer import calculate_bartlett_
sphericity
chi_square_value,p_value=calculate_bartlett_sphericity(pca_data)
chi_square_value, p_value
```

```
Out[69]: (23.012894934627802, 8.370237810650897e-07)
```

The value of 23.01 is an approximate chi-square, evaluated on a single degree of freedom. The p -value accompanying the test is extremely small, indicating that we can reject the null hypothesis of an identity matrix and infer the alternative hypothesis, that the correlation matrix is not an identity matrix, which gives us reason to proceed with the factor analysis. Even if the null hypothesis was not rejected here, it is certainly of no harm in trying the factor analysis anyhow. However, results may be dismal given there is little pairwise correlation among variables.

We can also obtain the **Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy**, which is a measure of the likely utility for conducting a factor analysis in the sense of providing an estimate of the proportion of variance in variables that might be due to underlying factors (Denis, 2020). Values of 0.70 are preferred, though even lower values (such as the 0.5 we obtain below) should not necessarily discount conducting the factor analysis. Both Bartlett’s and KMO should be used as guides only, not dogmatic criteria that should “halt” the factor analysis from taking place. We can compute the KMO in Python as follows:

```
from factor_analyzer.factor_analyzer import calculate_kmo
kmo_all,kmo_model=calculate_kmo(pca_data)

kmo_model
Out[71]: 0.5000000000000001
```

Now, let’s run the EFA on the data, requesting a single factor solution. We use the default method called **minimum residual (minres)** (see Comrey, 1962, for details) and code `n_factors=1` to request a single factor solution:

```

pip install FactorAnalyzer
from factor_analyzer import FactorAnalyzer
fa = FactorAnalyzer(rotation=None, method = 'minres', n_factors=1)
fa.fit(pca_data)
Out[225]:
FactorAnalyzer(bounds=(0.005, 1), impute='median', is_corr_
               matrix=False,
               method='minres', n_factors=1, rotation=None,
               rotation_kwargs={},
               use_smc=True)

```

To get the factor loadings, we request `fa.loadings_`:

```

fa.loadings_
Out[226]:
array([[ -0.98816761],
       [ 0.98816761]])

```

With two variables, the solution is not very meaningful, but if we were to interpret the above loadings at all, we could say that each variable loads very high onto the factor. This is hardly surprising, since the Pearson correlation coefficient between the variables is very high as well, equal to -0.976 . The **communalities** for each variable are shown. Each is equal to the squared loadings for each variable. That is, the first communality is computed as $(-0.98816761)(-0.98816761)$ while the second, $(0.98816761)(0.98816761)$:

```

fa.get_communalities()
Out[227]: array([0.97647522, 0.97647522])

```

As in our example on PCA, this factor analysis on two variables is rather contrived. You would rarely ever conduct a factor analysis on only two variables. Typically, factor analyses are conducted on data sets in which the number of variables is much larger. We ended the previous chapter with an example of PCA on the USA arrests data and we will do the same here with an EFA on the same data.

11.6 Exploratory Factor Analysis on USA Arrests Data

Recall the USA arrests data from the previous chapter. We print the first 10 cases using `data.head(10)`, where “10” indicates the top 10 cases:

```

data = pd.read_csv('usarrests.csv')
data.head(10)
Out[639]:

```

	Unnamed: 0	Murder	Assault	UrbanPop	Rape
0	Alabama	13.2	236	58	21.2
1	Alaska	10.0	263	48	44.5
2	Arizona	8.1	294	80	31.0
3	Arkansas	8.8	190	50	19.5

4	California	9.0	276	91	40.6
5	Colorado	7.9	204	78	38.7
6	Connecticut	3.3	110	77	11.1
7	Delaware	5.9	238	72	15.8
8	Florida	15.4	335	80	31.9
9	Georgia	17.4	211	60	25.8

Instead of a PCA as we did in the previous chapter, let's run a factor analysis on the variables murder, assault, urbanPop, and rape. To perform the factor analysis, we will use Python's **FactorAnalyzer** library:

```
pip install factor_analyzer
import pandas as pd
from factor_analyzer import FactorAnalyzer
```

As we did when we analyzed the PCA data, we first designate the variables on which we will subject to the factor analysis, as well as import `StandardScaler()`, which will allow us to standardize the data:

```
df = pd.DataFrame(data, columns=['Murder', 'Assault', 'UrbanPop', 'Rape'])
```

We see here that we have designated our dataframe having variables murder, assault, urbanPop, and rape. Let's now standardize our dataframe:

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(df)
```

```
Out[649]: StandardScaler(copy=True, with_mean=True, with_std=True)
```

We create the standardized data using `scaler.transform()`:

```
scaled_data = scaler.transform(df)
scaled_data

Out[650]:
array([[ 1.25517927,  0.79078716, -0.52619514, -0.00345116],
       [ 0.51301858,  1.11805959, -1.22406668,  2.50942392],
       [ 0.07236067,  1.49381682,  1.00912225,  1.05346626],
       [ 0.23470832,  0.23321191, -1.08449238, -0.18679398],
       [ 0.28109336,  1.2756352,  1.77678094,  2.08881393],
       [ 0.02597562,  0.40290872,  0.86954794,  1.88390137],
       [-1.04088037, -0.73648418,  0.79976079, -1.09272319],
       [-0.43787481,  0.81502956,  0.45082502, -0.58583422],
```

We first try the factor analysis without rotation (by specifying `rotation = None`) and using `fa.fit()` to run the factor analysis. For demonstration, we will attempt to extract four factors:

```
fa = FactorAnalyzer()
fa.set_params(n_factors=4, rotation=None)
fa.fit(scaled_data)
```

```
Out[654]:
FactorAnalyzer(bounds=(0.005, 1), impute='median', is_corr_
               matrix=False,
               method='minres', n_factors=4, rotation=None,
               rotation_kwargs={},
               use_smc=True)
```

We request the loadings via `fa.loadings_`:

```
fa.loadings_
Out[406]:
array([[0.84370394, -0.37474146, -0.07321271, 0.],
       [0.91937036, -0.10367227, 0.17283121, 0.],
       [0.33167926, 0.54884826, 0.06269644, 0.],
       [0.78479779, 0.29235871, -0.15025673, 0.]])
```

Now, recall that we requested the factor analysis to run on four variables, and hence, by default, it was only able to generate a maximum of three factors. To the contrary, recall that the equivalent PCA was, by definition, able to generate four components. Again, this is one big difference between PCA and EFA. In EFA, **commonality** among variables is a priority, which means for most versions of factor analysis, the technique will typically not generate as many factors as there are variables subjected to it (see Johnson and Wichern, 2007, p. 488, for an exception, which is a version of factor analysis basing itself directly on principal components). Each column in the above output is a presumed factor, and each number under it corresponds to the loading for the given variables of murder, assault, urbanpop, and rape. Hence, the first factor seems to be made up primarily of murder, assault, and rape, having loadings of 0.84, 0.91, and 0.78, respectively. The second and third factors are not defined very well having relatively small loadings (though an argument could be made for a second factor).

We can get the original (i.e., pre-EFA) eigenvalues from the correlation matrix, which will mirror those obtained in PCA:

```
ev, v = fa.get_eigenvalues()
ev
Out[375]: array([2.48024158, 0.98976515, 0.35656318, 0.17343009])
```

Next, we obtain the **communalities** for each variable across the factor solution:

```
fa.get_communalities()
Out[656]: array([0.85762759, 0.88586043, 0.41517638, 0.72395827])
```

There are four communalities because there are four observed variables (not the fact that we are extracting three factors). We now summarize the computation of the first communality, that of 0.85762759:

```
(0.84370394)**2 + (-0.37474146)**2 + (-0.07321271)**2
Out[661]: 0.8576276011199994
```

Note that the sum of squared loadings across the factor solution is equal to the computed communality. The specific variance is thus equal to $1 - 0.85762759$:

```
1-0.85762759
```

```
Out[662]: 0.14237241
```

The remaining communalities are computed in an analogous fashion. Let's now try a **varimax rotation**, where again we will obtain a three-factor solution. The idea of rotation here will be to rotate the loadings such that factors remain orthogonal. Varimax will attempt to maximize the variance in loadings for a given factor, which translates to boosting high loadings even higher and minimizing smaller loadings. Since the original axes of the factor analysis are in truth arbitrary, rotation in factor analysis is allowable, and, in a strong sense, the new axes have just as much "truth" to them as the original axes. What will determine ultimate justification regarding whether the rotation was successful or useful is whether it helps to make better sense of the factor solution, which should be informed by researcher judgment. We perform the rotation:

```
fa_varimax = FactorAnalyzer(rotation='varimax')
fa_varimax.fit(scaled_data)
```

```
Out[429]:
```

```
FactorAnalyzer(bounds=(0.005, 1), impute='median', is_corr_
               matrix=False,
               method='minres', n_factors=3, rotation='varimax',
               rotation_kwargs={}, use_smc=True)
```

```
fa_varimax.loadings_
```

```
Out[666]:
```

```
array([[0.91516486, 0.02762848, 0.13905951],
       [0.88036791, 0.32020407, -0.09100617],
       [0.05909459, 0.64142576, -0.0160376],
       [0.56292624, 0.59517677, 0.22986287]])
```

We can see that the varimax rotation had the effect of more or less making urbanpop (the third variable) pretty much irrelevant to the first factor, decreasing the loading from 0.33 in the original solution to only 0.05 in the rotation. However, the overall structure for the first factor remains pretty much intact, in that it is still made up of murder, assault, and rape. For the second factor, urbanpop and rape are relatively dominant in size. In both solutions, no case can be made for a third factor, as in both solutions, unrotated and rotated, the loadings are quite small. Hence, depending on one's theoretical stance or what one hoped to see from the solution, a case can be made for either a one-factor or two-factor solution (though the second factor does not look nearly as convincing as the first).

The communalities follow. We can see that urbanpop is the least relevant variable to the factor solution having a communality of only 0.41, while the other variables' communalities are relatively strong. That is, urbanpop is not explained as well by the factor solution as the other variables subjected to the analysis.

```
fa_varimax.get_communalities()
```

```
Out[668]: array([0.85762759, 0.88586043, 0.41517638, 0.72395827])
```

Review Exercises

1. What is **exploratory factor analysis (EFA)**? Why might a researcher want to perform one?
2. Why is **EFA** not equivalent to **PCA**? Explain two reasons why the two approaches are not the same and are used for different purposes.
3. Review and discuss the **common factor analysis model**. How is the model similar yet different from that of the regression model?
4. What does the **error term** indicate in the common factor analysis model?
5. What does it mean to say **factor analysis** is nothing more than an attempt to reproduce a covariance (or correlation) matrix? Why is this definition important to understand? What does this definition emphasize that other definitions may not?
6. Distinguish between an **observed** vs. **latent** variable. How are they different? Similar?
7. Discuss the nature of a **construct** in science. What is a construct exactly?
8. How does **measurement error** figure so prominently in science? Why can it be said to exist in virtually all variables?
9. What does it mean to say **factor loadings** are not **unique**, and why is this a major technical as well as philosophical pitfall to factor analysis?
10. Consider the **Breast Cancer Wisconsin** data from the **Machine Learning Repository**. Recall from the previous chapter that you conducted a PCA on this data. Attempt an **EFA** of the same variables and summarize your findings. How do they compare to the PCA you ran? Perform the EFA with and then without **varimax rotation**.